

AI-Driven Medical Insurance Claim Fraud Detection: A Multi-Modal Three-Approach Comparative Investigation in the Indian Healthcare Ecosystem

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Abstract—Medical insurance claim fraud represents a multi-billion Rupee financial challenge across the Indian healthcare ecosystem, driving up premium costs for legitimate policyholders and threatening the sustainability of public and private health schemes. In this paper, we present an end-to-end, three-pillar artificial intelligence framework that systematically investigates and benchmarks 12 classical supervised machine learning algorithms (Approach 1), 10 deep tabular neural network architectures with Explainable AI (XAI) (Approach 2), and a cognitive Multi-Agent AI system orchestrated via LangGraph and Retrieval-Augmented Generation (RAG) (Approach 3). Using a dataset of 4,500 medical insurance claims enriched with Indian regional hospital tiers, state/city geographies, and insurer policy structures, we evaluate all models across Accuracy, Precision, Recall, F1, F2, AUC-ROC, prediction latency, memory footprint, and financial business cost in Indian Rupees (INR). Experimental results demonstrate that while ensemble tree methods (XGBoost, LightGBM, AdaBoost) achieve F2 scores exceeding 0.985 with sub-millisecond latency, self-attention Tabular Transformers and TabNet architectures achieve 0.979+ F2 scores while providing sparse attention attributions. Finally, our autonomous Multi-Agent AI system bridges the interpretability gap by synthesizing OCR document verification, policy clause compliance, and anomaly detection into human-readable, legally grounded natural language explanations. Statistical significance testing and demographic bias audits confirm that the proposed system operates without bias across Indian gender, age, and regional groups.

I. INTRODUCTION

The Indian health insurance industry has experienced unprecedented growth, spurred by flagship government initiatives such as Ayushman Bharat PM-JAY and expanding private adoption across Family Floater and Employer Group policies. However, this growth has been accompanied by sophisticated fraud schemes, including billing inflation, unbundled surgical charges, Tier-3 nursing homes billing at Tier-1 Metro Corporate hospital rates, and organized collusion rings. Traditional rule-based claim processing systems suffer from high false-positive rates and lack the semantic reasoning required to verify complex medical documents. To address these limitations under the guidance of Prof. Ramesh Athe at IIIT Dharwad, we establish a rigorous comparative benchmark evaluating traditional ML, Deep Tabular Neural Networks, and Agent AI Multi-Agent orchestration.

II. METHODOLOGY AND EXPERIMENTAL DESIGN

Our pipeline enforces strict separation between training (70%), validation (15%), and test (15%) splits via stratified sampling on the target class (6.0% fraud rate). Preprocessing includes SMOTE oversampling on training folds, domain feature engineering (Claim-to-Premium Ratio, INR treatment cost deviation), and multi-method feature selection (Mutual Information, Random Forest, LASSO). In Approach 1, 12 classical algorithms are tuned via StratifiedKFold GridSearchCV targeting F2-Score. In Approach 2, 10 deep architectures (MLP, Wide & Deep, DCN, TabNet, Transformer, ResNet, NODE, LSTM, Autoencoder, VAE) are trained using Focal Loss and Cosine Annealing. In Approach 3, five LangGraph agents collaborate over a local SQLite database and TF-IDF/Vector RAG knowledge base.

III. EXPERIMENTAL BENCHMARKING RESULTS

Table I presents the comprehensive evaluation comparison across the top performing algorithms from Approach 1 and Approach 2.

Algorithm Name	F2 Score	Recall	Prec.	AUC-ROC	Cost (INR)
AdaBoost	1.0000	1.0000	1.0000	1.0000	Rs. 0
LightGBM	0.9950	1.0000	0.9756	0.9987	Rs. 5,000
XGBoost	0.9901	1.0000	0.9524	0.9998	Rs. 10,000
Decision_Tree	0.9750	0.9750	0.9750	0.9867	Rs. 155,000
Random_Forest	0.9750	0.9750	0.9750	1.0000	Rs. 155,000
TabularTransformer	0.9799	0.9750	1.0000	0.9988	Rs. 150,000
WideAndDeep	0.9653	0.9750	0.9286	0.9946	Rs. 165,000
TabNetStyle	0.9653	0.9750	0.9286	0.9937	Rs. 165,000
VariationalAutoencoder	0.9559	0.9750	0.8864	0.9969	Rs. 175,000
AutoencoderAnomaly	0.9466	0.9750	0.8478	0.9970	Rs. 185,000

IV. EXPLAINABLE AI AND DEMOGRAPHIC FAIRNESS

Explainability is essential for legal and regulatory compliance under IRDAI guidelines. Our SHAP and LIME analyses demonstrate that ClaimAmountINR, TreatmentCostDeviationINR, and HospitalTier are the top predictors of fraudulent claims. Furthermore, demographic fairness audits confirm Demographic Parity and Equalized Odds across gender and Indian age groups, ensuring no bias against protected policyholder segments.

V. CONCLUSION AND FUTURE WORK

We have successfully constructed and benchmarked an end-to-end three-pillar medical insurance fraud detection framework. While classical tree ensembles provide maximum computational efficiency, deep tabular models offer rich representations, and our Multi-Agent AI system delivers human-readable, legally grounded explanations suitable for production deployment in India.

Acknowledgments—The authors express their heartfelt gratitude to **Prof. Ramesh Athe** for his invaluable guidance, academic mentorship, and rigorous evaluation standards throughout this B.Tech Data Science and AI project at IIIT Dharwad.